CSCI 5521: Machine Learning Fundamentals (Spring 2022)[1]

Midterm Exam

Due on Gradescope by 1pm, Mar 18

Instructions:

- This test has 4 questions, 100 points + 3 points (extra credits), on 4 pages.
- **For full credit, show how you arrive at your answers**.
- You have **24 hours** to complete and submit this test to gradescope.

1. (**32 points**) In I-IV, select the correct option(s) (it is not necessary to explain).

I. Select all the option(s) that correspond to supervised-learning algorithms:
(a) k-Means
(b) Linear discriminant analysis
(c) Linear discrimination
(d) Linear regression

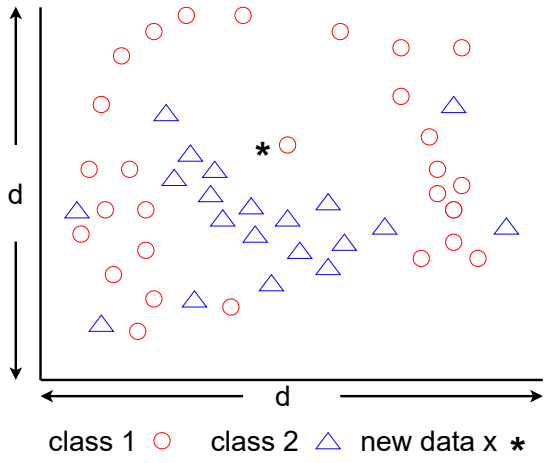
II. What could we do to reduce overfitting? Select all the option(s) that apply:
(a) Add more training data and keep testing data the same
(b) Add more testing data and keep training data the same
(c) Change to a less complex model (e.g., a model with less parameters)
(d) Use mixture of Gaussians instead of k-means for clustering

III. Select all the option(s) that correspond to dimensionality reduction algorithms:
(a) Linear discriminant analysis (b) Mixture of Gaussians
(c) Principal component analysis (d) Expectation maximization

IV. Select all the true statement(s) below.:
(a) Both MLE and MAP are parametric.
(b) Both MLE and MAP consider prior information.
(c) Expectation Maximization finds local optimum instead of global ones.
(d) Gradient descent finds global optimum instead of local ones.

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2. (**22 points**) Given a set of data points $\{x^t\}$ each shown in the figure, find the label of a new data point x using different non-parametric estimators / classifications as specified below.



(a) Write down the label of the new data point x with k nearest neighbor estimator when $k = 1$. Briefly explain the reason.

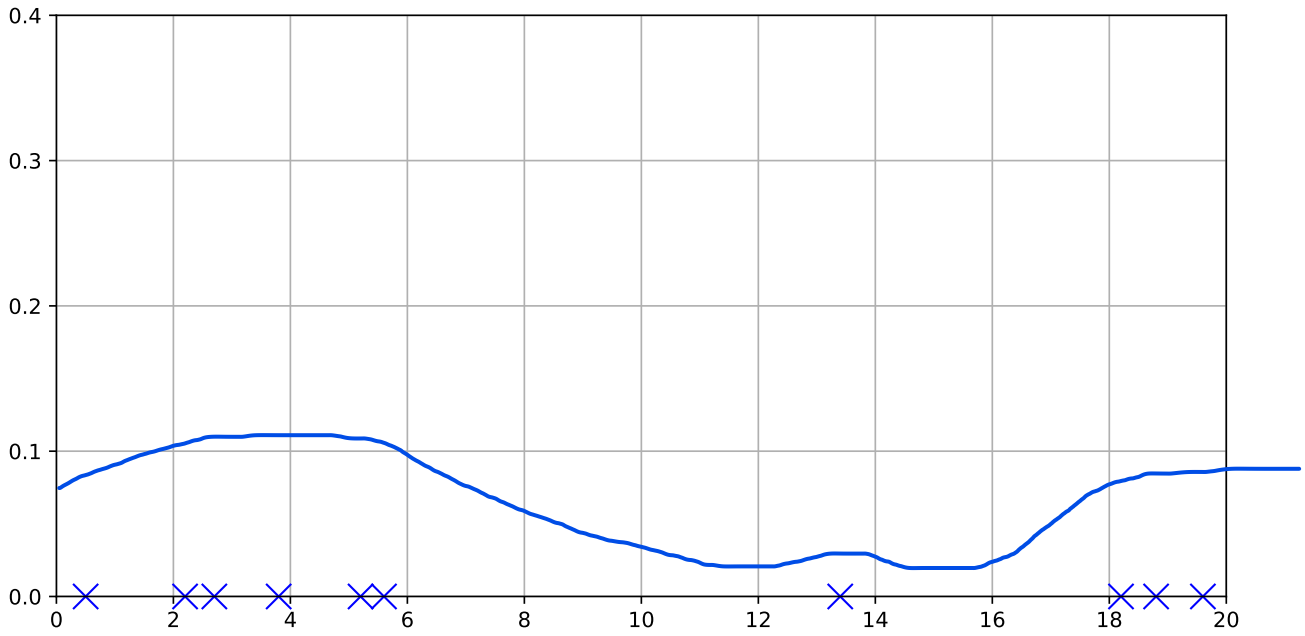
(b) Write down the label of the new data point x with k nearest neighbor estimator when $k = 20$. Briefly explain the reason.

(c) Assume a uniform kernel function:

$$K = \begin{cases} \frac{1}{\pi d^2}, & ||x - x^t||_2 \leq d \\ 0, & \text{otherwise} \end{cases}$$

Write down the label of the new data point x with kernel estimator. Justify your answer.

3. (**24 points**) Answer the following questions about non-parametric density estimator:



(a) Draw a histogram estimator (start from origin) using $h = 4.0$ for the following 10 training data points in $\mathbb{R}$: 0.5, 2.2, 2.7, 3.8, 5.2, 5.6, 13.4, 18.2, 18.8, 19.6

(b) Given a test data point $x = 6.0$, what is the predicted density $p(x)$ for the data point? Show your work for full credits.

(c) List one possible approach to get a smoother density estimate. Draw an approximate curve when the kernel is used. You do not need to show the calculation.

4. (**22+3 points**) Given a set of N iid random variables $\mathcal{X} = x_1, ...x_N$ and $\mathcal{Z} \in \mathbb{R}^{N \times K}$, the likelihood function is given by:

$$\mathcal{L}(\pi, \theta | \mathcal{X}, \mathcal{Z}) = p(\mathcal{X}, \mathcal{Z} | \pi, \theta) = log \prod_{n=1}^{N} \prod_{k=1}^{K} \pi_k^{z_{nk}} Bern(x_n | \theta_k)^{z_{nk}} \tag{1}$$

such that $\sum_{k=1}^{K} \pi_k = 1$, $\sum_{n=1}^{N} \sum_{k=1}^{K} z_{nk} = N$ and $Bern(x|\theta_k) = \theta_k^x (1 - \theta_k)^{1-x}$.

(a) Derive the MLE of θ_k.

(b) (extra credits, 3 points) Derive the MLE of π_k. Hint: there is a constraint on π_k.